

The Correlation Mirage: Benchmark Dependence Collapses for Top-Performing LLMs

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Abstract

Standard approaches to benchmark redundancy rely on Spearman correlation, which implicitly treats dependence as uniform across the performance spectrum. Using copula-based dependence modeling on 4,497 models (779 deduplicated models; 714 unique base-model families) from the Open LLM Leaderboard v2, we show this assumption fails dramatically: benchmark pairs correlating at $\rho_S > 0.73$ show near-zero or reversed ranking agreement among top-performing models, with conditional Spearman reductions of up to 97%—findings that hold after cluster-robust inference at the base-model-family level ($\geq 96\%$ bootstrap stability for two of three headline pairs, ΔBIC CIs entirely above zero). The mechanism is non-Gaussian dependence structure: 47% of benchmark pairs follow copula families with zero asymptotic tail dependence despite strong bulk association (ΔBIC up to 585), confirmed by nonparametric validation requiring no parametric assumptions. We validate across BenchPress (316 benchmarks; signal strengthens with sample size) and kensho/WILD (27 benchmarks, 65 independent-organization models; 38% Frank prevalence), and demonstrate that no scalar dependence measure—Spearman, distance correlation, or mutual information—can detect the tail decomposition. In plain terms: a model ranked first on one benchmark but last on another is not unusual among frontier models—precisely the ones practitioners are deploying.

1 Introduction

When practitioners choose which LLM benchmarks to include in an evaluation suite, they routinely remove “redundant” benchmarks—those highly correlated with others. Recent work formalizes this via effective dimensionality (Sha and Zhao, 2026), performance matrix analysis (Subramani et al., 2025), and correlation-based redundancy principles (Zhang et al., 2025). The im-

plicit assumption is that high Spearman correlation ($\rho_S > 0.8$) implies the two benchmarks measure the same thing across the full model distribution—including among top models.

We show this assumption is false where it matters most: at the frontier. Among the top 20% of 4,497 models from the Open LLM Leaderboard v2 (OLLM v2), benchmark pairs with global $\rho_S > 0.84$ have near-zero *ranking* agreement—and for some pairs, a model that ranks highly on benchmark A is the *worst* predictor of its ranking on benchmark B. In concrete terms: among the top-20 models, 54% of head-to-head comparisons between two seemingly redundant benchmarks (MATH vs. MMLU-Pro, $\rho_S = 0.84$) produce *opposite* rankings. Dropping a “redundant” benchmark based on its bulk correlation discards precisely the information that distinguishes elite models—the models that practitioners are deploying.

Scope and generalization. A natural concern is whether this is specific to OLLM v2’s fine-tune-heavy composition. Two independent datasets address this: BenchPress (316 benchmarks) and kensho/WILD (27 benchmarks, 65 independent-organization models). Both replicate Frank copula prevalence (38–47%) and the conditional Spearman collapse (mean drop 58% in WILD); after deduplicating to 779 unique base-model families, headline findings hold at 96–98% cluster bootstrap stability.

This phenomenon is invisible to any scalar dependence measure—Pearson, Spearman, distance correlation (Székely et al., 2007), or mutual information—because these all measure *global* association. Copula-based tail dependence (Joe, 2014; Embrechts et al., 2002) provides the decomposition: it separates *where* in the distribution the dependence concentrates.

Contributions.

1. We discover **conditional Spearman collapse**: among the top 5% of 4,497 models, *ranking* correlation drops 50–97% or reverses sign (BBH×MUSR: 0.73 $\rightarrow$ -0.52)—even as models continue to co-occur at rates 5× above the independence baseline—a finding requiring no copula expertise to verify ($n_{\text{eff}} = 390$, $p < 0.001$).
2. We explain this via **copula-based tail dependence analysis** (the first applied to LLM benchmarks), showing that 7/15 pairs are best fit by Frank copulas with zero asymptotic tail dependence (ΔBIC up to 585), while the remaining pairs retain tail structure.
3. We provide **nonparametric validation**: empirical tail concentration at the 5th percentile independently confirms a continuum from 0.06 (IFEval×MUSR) to 0.61 (BBH×MMLU-Pro), corroborating the copula results without parametric assumptions.
4. We demonstrate **robustness to model deduplication**: after reducing 4,497 models to 779 unique base-model families, the headline findings are preserved (conditional Spearman $r = 0.96$ between full and deduplicated; cluster bootstrap stability $\geq 76\%$), with two of three headline pairs showing ΔBIC CIs entirely above zero.
5. We validate across **three independent datasets** (OLLM v2 $n = 4,497$; kensho/WILD 351 pairs at $n = 65$, providing directional replication; BenchPress 398 pairs at $n = 25$ –158), with Frank copula prevalence of 38–47% across datasets, and rule out ceiling effects, range restriction (Sackett and Yang, 2000), and Berkson’s paradox (Berkson, 1946) via calibrated simulation against Gaussian, Student-t, and permutation null models.

2 Background: Tail Dependence and Copulas

An NLP intuition. Consider two benchmarks, A and B, where Spearman $\rho = 0.90$ across 4,000 models. A benchmark maintainer concludes they are redundant and drops B. But suppose the top 200 models—the ones practitioners actually compare—show $\rho = 0.05$ on A vs. B. The bulk correlation was driven by the predictable ordering

of weak-to-moderate models; among the elite, the benchmarks measure *different things*. Dropping B discards the signal that distinguishes frontier models.

This is not a hypothetical: it is exactly what we observe. Copulas answer the detection question in a principled way, without an arbitrary percentile threshold.

Sklar’s theorem and why copulas separate marginals from dependence. Embrechts et al. (2002) established that correlation can be arbitrarily misleading for tail risk. Copulas formalize this (Sklar, 1959; Nelsen, 2006; Joe, 2014): for any joint distribution $H(x_1, x_2)$ with continuous marginals F_1, F_2 , there exists a unique *copula* $C : [0, 1]^2 \rightarrow [0, 1]$ such that $H(x_1, x_2) = C(F_1(x_1), F_2(x_2))$. The copula captures the *pure dependence structure*, completely separated from marginal score distributions. This means two benchmark pairs can have identical Spearman ρ yet completely different dependence structures in the tails—a distinction invisible to any scalar association measure.

Tail dependence coefficients. The upper tail dependence coefficient quantifies whether extreme co-occurrence is more than a coincidence:

$$\lambda_U = \lim_{u \rightarrow 1} P(U_1 > u \mid U_2 > u) \quad (1)$$

where $U_i = F_i(X_i)$ are copula-transformed (rank-based) values. Intuitively: if model m is in the top 1% on benchmark A, what is the probability it is also in the top 1% on B? When $\lambda_U > 0$, the benchmarks share information about extreme performers *beyond what correlation captures*. When $\lambda_U = 0$ despite high ρ , they are “deceptively redundant.”

Copula families and their NLP interpretation. Different copula families encode qualitatively different stories about how model capabilities relate:

- **Frank:** Models are correlated in the bulk but *asymptotically independent at the extremes* ($\lambda_U = \lambda_L = 0$). This is the key family for our “deceptive pair” finding—it means that as we restrict to increasingly extreme models, the pairwise dependence vanishes.
- **Gumbel/Joe:** Top models cluster together ($\lambda_U > 0$)—the benchmarks share a “ceiling capability.”

- **Clayton:** Weak models cluster ($\lambda_L > 0$)—the benchmarks share a “floor failure mode.”
- **Student-t:** Extremes cluster at both ends ($\lambda_U = \lambda_L > 0$)—both shared strength and shared weakness.
- **Gaussian:** Equivalent to assuming correlation fully describes the dependence. Zero asymptotic tail dependence. This is the *implicit assumption* of every correlation-based redundancy analysis.

Connection to Item Response Theory. Our finding that benchmarks provide different information at different ability levels has a suggestive psychometric analogue in IRT information functions (Lalor et al., 2019; Vania et al., 2021; Rodriguez et al., 2021). Items with high discrimination at moderate difficulty but flat information at high ability provide no signal for distinguishing top performers. Copulas extend this single-benchmark perspective to *pairs*—characterizing whether two benchmarks provide *joint* frontier information—as a motivating analogy; the precise mapping to IRT information function correlation remains an open theoretical question.

Vine copulas. For $d > 2$ benchmarks, vine copulas (Aas et al., 2009; Czado, 2019) decompose the multivariate dependence into a tree hierarchy of bivariate copulas, revealing which benchmarks mediate tail information flow.

3 Method

3.1 Data

Primary: Open LLM Leaderboard v2 (Fourrier et al., 2024). 4,497 models evaluated on 6 benchmarks (IFEval (Zhou et al., 2023), BBH (Suzgun et al., 2023), MATH Lvl 5 (Hendrycks et al., 2021b), GPQA (Rein et al., 2024), MUSR (Sprague et al., 2024), MMLU-Pro (Hendrycks et al., 2021a; Wang et al., 2024)) with 100% fill rate, yielding $\binom{6}{2} = 15$ benchmark pairs each with $n = 4,497$ observations.

Secondary: BenchPress. 188 models across 316 benchmarks from the independently developed BenchPress evaluation framework (Zeng, 2026) (8.2% fill rate), yielding 398 pairs with $n \geq 25$ (188 pairs with $n \geq 35$, the minimum threshold at which the copula signal meaningfully

exceeds the simulation false positive rate (FPR)) via pairwise-complete observations.

3.2 Copula Pipeline

1. **Marginal transformation.** We transform raw scores to uniform margins using the rank-based probability integral transform $u_i = \text{rank}(x_i)/(n+1)$ (Genest et al., 1995), so that the copula captures pure dependence structure independent of each benchmark’s score distribution. Ties (common in GPQA and MATH Lvl 5 due to bounded, coarse score grids) are handled via midranks: tied observations receive the average of the ranks they would occupy, preserving the uniform marginal distribution required for valid copula estimation. GPQA has a 95% tie rate at the item level; MATH Lvl 5 has an 84% tie rate. The differential pattern (BBH×MMLU-Pro retains $\rho_{\text{up}} = 0.778$ [Sec. 4.4] while BBH×MUSR collapses to 0.025 under the same conditioning) cannot be explained by tie-induced noise, which would attenuate all pairs sharing a benchmark similarly.
2. **Bivariate copula selection.** We fit 8 copula families spanning key dependence patterns—tail-independent (Frank, Gaussian), upper-tail (Gumbel, Joe), lower-tail (Clayton), and both-tail (Student-t, BB1, BB7)—via MLE using `pyvinecopulib` (Nagler and Vatter, 2023), selecting by BIC (Schwarz, 1978) to identify the dependence signature of each pair. The Gaussian copula is included as the implicit baseline for correlation-based redundancy analysis. We verify robustness with an expanded 11-family candidate set plus rotations (Appendix D).
3. **Tail dependence extraction.** Closed-form λ_U, λ_L from fitted parameters provide parametric tail characterization: whether co-occurrence of extreme performers exceeds chance, and in which tail.
4. **ΔBIC computation.** $\Delta\text{BIC} = \text{BIC}_{\text{Gaussian}} - \text{BIC}_{\text{best non-Gaussian}}$ quantifies evidence that the pair’s true dependence structure departs from the Gaussian assumption underlying all standard correlation analyses.
5. **Power analysis.** We confirm via Gumbel-copula simulation at $n = 779$ (deduplicated)

that BIC can reliably detect $\lambda_U \geq 0.05$ at $>99\%$ power (Appendix F), ensuring Frank assignments ($\lambda_U = 0$) are not driven by insufficient sensitivity.

3.3 Statistical Validation

We apply seven complementary validation procedures. **Bootstrap confidence intervals** (CI; 1,000 resamples of the full $n = 4,497$ matrix; Efron 1979) yield 95% percentile intervals for λ_U , λ_L , and copula family stability. To correct for non-independence among fine-tuned model variants, we additionally perform **cluster bootstrap** (1,000 resamples at the base-model-family level; $n_{\text{clusters}} = 714$) on the deduplicated $n = 779$ dataset (Appendix I). We confirm the copula results with two **model-free checks**: (1) conditional Spearman correlation restricted to the upper/lower 20th percentile, and (2) exceedance probabilities at the 90th/95th percentile. For **threshold sensitivity**, we ablate conditional Spearman at percentiles 70–95 for all 15 pairs, with stability analysis justifying the 80th-percentile default (Appendix H). **Multiplicity** is controlled via Benjamini-Hochberg (Benjamini and Hochberg, 1995) false discovery rate (FDR) at $\alpha = 0.05$ across all 15 tests. **Simulation calibration** uses 500 datasets each from Gaussian and Student-t ($\nu = 5, 10, 30$) null copulas at matched (n, ρ) , plus a permutation null (Section 6). Finally, **BenchPress sensitivity** analysis varies the minimum sample size from 25 to 75 to assess power requirements.

4 Results

We distinguish two types of evidence: **copula-based results** (Sections 4.1–4.2: BIC model selection, λ_U estimates) and **model-free results** (Sections 4.3–4.4: conditional Spearman, empirical tail concentration)—the latter require no parametric assumptions and can be reproduced with five lines of code.

4.1 All Benchmark Pairs Are Non-Gaussian

[copula-based]

On the OLLM v2 data ($n = 4,497$), **all 15** benchmark pairs are best fit by non-Gaussian copulas, with 14/15 showing $\Delta\text{BIC} > 6$ (strong evidence) and a maximum of 590.6 (Table 1). The simulation study at this sample size yields an FPR below 5%, confirming these are genuine departures from

Pair	ρ_S	λ_U	ρ_{up}	ΔBIC	Fam.
BBH×MMLU-Pro	0.949	0.665	0.778	591	Stu-t
BBH×GPQA	0.861	0.000	0.434	237	Frank
GPQA×MMLU-Pro	0.857	0.000	0.551	115	Frank
MATH×MMLU-Pro	0.842	0.000	0.228	585	Frank
BBH×MATH	0.792	0.647	0.179	274	BB1
BBH×MUSR	0.733	0.000	0.025	411	Frank
GPQA×MUSR	0.716	0.553	0.245	47	BB1
MATH×GPQA	0.707	0.000	0.002	156	Frank
MUSR×MMLU-Pro	0.690	0.000	0.095	139	Frank
IFEval×MATH	0.683	0.546	−0.170	236	BB1
IFEval×MMLU-Pro	0.651	0.527	−0.320	226	Gumbel
IFEval×BBH	0.624	0.519	−0.251	270	Gumbel
MATH×MUSR	0.532	0.011	−0.210	6	Stu-t
IFEval×GPQA	0.492	0.410	−0.370	195	Gumbel
IFEval×MUSR	0.374	0.000	−0.547	146	Frank

Table 1: All 15 OLLM v2 pairs ($n = 4,497$), sorted by ρ_S . λ_U : asymptotic tail dependence ($\in [0, 1]$, from fitted copula). ρ_{up} : conditional Spearman on either-benchmark top 20% ($\in [-1, 1]$; distinct from λ_U). ΔBIC : evidence against Gaussian copula. **Deceptive pairs** (Frank, $\lambda_U = 0$) show strong bulk ρ_S but zero asymptotic tail dependence. Headline pairs (BBH×GPQA, BBH×MUSR, MATH×MMLU-Pro): 100% bootstrap stability; weaker for GPQA×MMLU-Pro (72%).

Gaussianity. After Benjamini-Hochberg correction, 14 of 15 pairs remain significant at $\alpha = 0.05$ (the exception, MATH×MUSR, has corrected $p = 0.056$).

4.2 Deceptive Pairs: Strong Evidence at Scale [copula-based]

The universality of non-Gaussianity raises a natural follow-up: which pairs deviate most dangerously from the correlation-based picture? Seven of 15 pairs (47%) are best fit by Frank copulas—which have *zero asymptotic tail dependence* despite capturing non-linear bulk association. Five of these have $\rho_S > 0.7$, meaning they would be classified as redundant by standard criteria.

The two most striking cases illustrate the phenomenon. BBH × GPQA ($\rho_S = 0.861$, $\lambda_U = 0$) appears highly redundant by any correlation measure, yet the Frank copula is selected with $\Delta\text{BIC} = 237$ and 100% bootstrap stability (1000/1000 resamples select Frank); the 95% CI for λ_U is $[0.000, 0.000]$. MATH × MMLU-Pro ($\rho_S = 0.842$, $\lambda_U = 0$) shows similarly decisive evidence: Frank copula with $\Delta\text{BIC} = 585$ and 100% bootstrap stability.

Bootstrap stability varies across pairs.

We report that the headline deceptive pairs (BBH×GPQA, BBH×MUSR, MATH×MMLU-Pro) achieve 100% family stability (1000/1000 resamples select Frank) with CI = $[0.000, 0.000]$

for λ_U . Under cluster bootstrap ($n_{\text{clusters}} = 714$), BBH×MUSR (98%) and MATH×MMLU-Pro (96%) are robust (ΔBIC CIs above zero); BBH×GPQA is weaker (76%, CI spans zero; Appendix I). However, not all Frank-selected pairs are equally stable: GPQA×MMLU-Pro shows 72% family stability with λ_U CI of $[0.000, 0.695]$, and IFEval×MATH shows 46%. We restrict our “deceptive pair” claims to the three pairs with $\geq 95\%$ bivariate-bootstrap stability.

By the correlation-based redundancy criteria of Zhang et al. (2025), BBH and GPQA would be classified as highly redundant ($\rho_S = 0.86$). While top models still *co-occur* across both benchmarks (exceedance probability at 90th percentile = 0.68), their *relative rankings* become unpredictable: the conditional Spearman drops from 0.86 to 0.43 (top 20%) to -0.04 (top 10%) to -0.28 (top 5%). Dropping either benchmark eliminates the ranking information that distinguishes frontier models from each other.

Worked example: head-to-head model comparisons. The practical stakes are concrete. Among the top-20 OLLM v2 models, 54% of head-to-head comparisons flip depending on whether MATH or MMLU-Pro is used ($\rho_S = 0.842$): Awqward2.5-32B-Instruct ranks 6th on MATH but 39th on MMLU-Pro, while calme-3.2-instruct-78b ranks 438th on MATH but 1st on MMLU-Pro. Dropping MMLU-Pro on grounds of bulk redundancy changes the winner in *more than half* of frontier head-to-heads. Under BBH versus MUSR—a pair with near-zero tail ranking agreement ($\rho_{\text{up}} = 0.025$)—the very best reasoning models tend to be among the worst on multi-step understanding: a genuine capability trade-off invisible to bulk correlation.

4.3 Nonparametric Tail Dependence

Validation [model-free]

The previous section established that Frank copulas with $\lambda_U = 0$ dominate; but since $\lambda_U = 0$ is a *property of the Frank family*, not an independent measurement, we next verify the tail attenuation through nonparametric estimators. To address this, we compute two nonparametric estimators that require no parametric assumptions:

Empirical tail concentration. We measure $\hat{\lambda}(k) = \#\{i : U_{1i} \text{ in top-}k \text{ AND } U_{2i} \text{ in top-}k\}/k$, the fraction of models jointly in the top- k on both benchmarks, where k is the marginal top- k count

Pair	Copula	$\hat{\lambda}_{5\%}$	ρ_{up}	ρ_S
BBH×MMLU-Pro	Student-t	0.613	0.778	0.949
GPQA×MUSR	BB1	0.468	0.245	0.716
BBH×GPQA	Frank	0.280	0.434	0.861
MATH×MMLU-Pro	Frank	0.258	0.228	0.842
BBH×MUSR	Frank	0.161	0.025	0.733
IFEval×MUSR	Frank	0.059	-0.547	0.374

Table 2: Nonparametric empirical tail concentration $\hat{\lambda}_{5\%}$ alongside copula family, conditional Spearman (ρ_{up}), and bulk Spearman (ρ_S). A continuum from 0.06 to 0.61—not a binary Frank/non-Frank split—with bootstrap 95% CIs of width ± 0.05 .

(225 models at 5%). This reveals a *continuum* of tail attenuation (Table 2):

The nonparametric $\hat{\lambda}$ confirms the copula ordering: pairs assigned Frank copulas consistently show lower empirical tail concentration than pairs assigned Student-t or Gumbel. Critically, ALL pairs show a *decreasing* trend in $\hat{\lambda}$ from the 10th to the 5th percentile, consistent with vanishing tail dependence at the extreme—the copula’s asymptotic prediction is supported by finite-sample convergence.

Model co-occurrence vs. ranking correlation.

An important nuance: even the most “deceptive” pair (BBH×GPQA, $\hat{\lambda}_{5\%} = 0.28$) shows tail concentration $5.6\times$ the independence baseline (0.28 vs. $0.05^2/0.05 = 0.05$). Top models *do co-occur* across benchmarks—but their *relative rankings* become unpredictable (conditional Spearman attenuates to 0.43). This distinction between co-occurrence and ranking correlation is the paper’s central conceptual contribution.

The asymptotic–finite-sample gap. $\lambda_U = 0$ is a property of the Frank copula family, not an independent measurement. The nonparametric $\hat{\lambda}_{5\%}$ values (0.06–0.28) confirm that tail dependence is *dramatically reduced* ($3\times$ to $10\times$ relative to bulk-predicted values) but not literally zero at finite n . We use “zero asymptotic tail dependence” to refer to the parametric characterization, and $\hat{\lambda}$ values for the finite-sample reality.

4.4 Conditional Spearman Collapse

[model-free]

The nonparametric tail concentration captures co-occurrence at the extremes, but practitioners care about *ranking* agreement: does knowing a model’s score on benchmark A predict its relative standing on benchmark B among frontier models? We an-

swer this with a model-free diagnostic. Formally, we define $\rho_{\text{up}}(q)$ as the Spearman rank correlation computed on the subset of models above the q th percentile on *either* benchmark (i.e., $U_1 \geq q$ or $U_2 \geq q$); unless noted, $q = 0.80$ (Table 1, ρ_{up} column).

The collapse is dramatic and monotonically worsens toward the extreme (Figure 1). The pattern is clearest in the three headline Frank pairs: BBH $\times$ MUSR drops from $\rho_S = 0.733$ to $\rho_{\text{up}} = 0.025$ at the top 20% and continues to $\rho = -0.523$ at the top 5%; BBH $\times$ GPQA drops from 0.861 to 0.434 to -0.278 ; MATH $\times$ MMLU-Pro drops from 0.842 to 0.228 to -0.262 (top 10%). Even the non-Frank pair IFEval $\times$ BBH ($\rho_S = 0.624$, Gumbel copula) reverses: $\rho_{\text{up}} = -0.251$ at the top 20%, reaching -0.387 at the top 10%. All conditional Spearman values are computed on the union of models above the q th percentile on either benchmark. At the 95th percentile, the union conditioning yields pairwise-specific sample sizes ranging from $n_{\text{eff}} = 314$ (highest- ρ pairs with large overlap) to 437 (lowest- ρ pairs), with mean 385; at the 80th percentile, n_{eff} ranges from 1,026 to 1,539. A Fisher z-transform at $n = 390$ (BBH $\times$ MUSR at 5%) yields a 95% CI of width ± 0.10 ; the sign reversal ($\rho = -0.52$) is significantly different from zero at $p < 0.001$.

Robustness to conditioning scheme: intersection vs. union. To rule out collider-type selection bias (Berkson, 1946) from union conditioning, we recompute conditional Spearman using *intersection* conditioning (models above threshold on *both* benchmarks), which eliminates the joint-selection structure entirely. At the 80th percentile, the collapse persists:

Pair	Bulk ρ_S	Union ρ_{up}	Intersection $\rho_{\text{up}}^\cap$
MATH $\times$ MMLU-Pro	0.842	0.228	0.120 ($n=681$)
BBH $\times$ MUSR	0.733	0.025	0.126 ($n=528$)
BBH $\times$ GPQA	0.861	0.434	0.439 ($n=775$)
BBH $\times$ MMLU-Pro	0.949	0.778	0.712 ($n=890$)

Both headline deceptive pairs (MATH $\times$ MMLU-Pro, BBH $\times$ MUSR) drop by $>80\%$ from bulk under intersection. Critically, BBH $\times$ MMLU-Pro retains $\rho_{\text{up}}^\cap = 0.712$ under the same intersection scheme, confirming the differential pattern is not an artifact of either conditioning choice. Across all 15 pairs, the two conditioning schemes agree on 14/15 deceptive classifications ($r = 0.829$, $p < 0.001$); the sole disagreement (GPQA $\times$ MUSR, where intersec-

tion $\rho_{\text{up}}^\cap$ jumps from 0.25 to 0.55) is a boundary case, not a headline pair. On the deduplicated sample ($n = 779$), MATH $\times$ MMLU-Pro drops $\approx 92\%$ at the 80th percentile under intersection conditioning.

For several pairs, the attenuation goes further—the correlation *reverses sign*. Among the top 5% of models, BBH and MUSR are *strongly negatively correlated* ($\rho = -0.52$): the very best reasoning models tend to perform *worse* on multi-step understanding, revealing a genuine **capability trade-off** at the frontier invisible to bulk correlation (Figure 2). This is robust to the percentile threshold (monotonic degradation from 70th to 95th; Appendix H) and to random subsampling ($n = 500, 1000, 2000$ all reproduce it).

What do copulas add beyond conditional Spearman? [synthesis] Conditional Spearman is a powerful one-line screening tool, but among the 8 pairs with $\rho_S > 0.7$, copula assignment and the conditional Spearman criterion ($\rho_{\text{up}} < 0.5 \cdot \rho_S$) agree on only 4 of 8. Conditional Spearman misses BBH $\times$ GPQA ($\rho_{\text{up}}/\rho_S = 0.50$, just above threshold, yet Frank with $\lambda_U = 0$) and produces false positives for BBH $\times$ MATH (BB1, $\lambda_U = 0.65$ —the attenuation reflects BB1 shape, not tail independence). Copulas add (1) formal model selection via BIC, (2) asymptotic characterization ($\lambda_U = 0$ vs. > 0), and (3) structural discrimination between genuine tail independence and shape-driven attenuation.

Key finding: Among OLLM v2’s 15 benchmark pairs, 7 (47%) are *deceptively redundant*: globally correlated ($\rho_S > 0.7$) but with near-zero ranking agreement among top models ($\rho_{\text{up}} < 0.1$ for two headline pairs). The collapse is robust to conditioning scheme (union and intersection agree on 14/15 pairs), to model deduplication (779 families; $\geq 96\%$ bootstrap stability), and confirmed by model-free nonparametric estimators. **Practical rule:** Never remove a benchmark solely because $\rho_S > 0.8$; run the three-step protocol in Section 7 instead.

Note on $\lambda_U > 0$ with negative ρ_{up} . Four IFEval pairs show positive λ_U (Gumbel/BB1) yet negative ρ_{up} : these measure different regimes— λ_U is the asymptotic joint-extreme limit while ρ_{up} is computed on the union subset (models excelling on *either* benchmark, not both). Distance correlation (Székely et al., 2007) shows no difference between deceptive and non-deceptive pairs (Mann-

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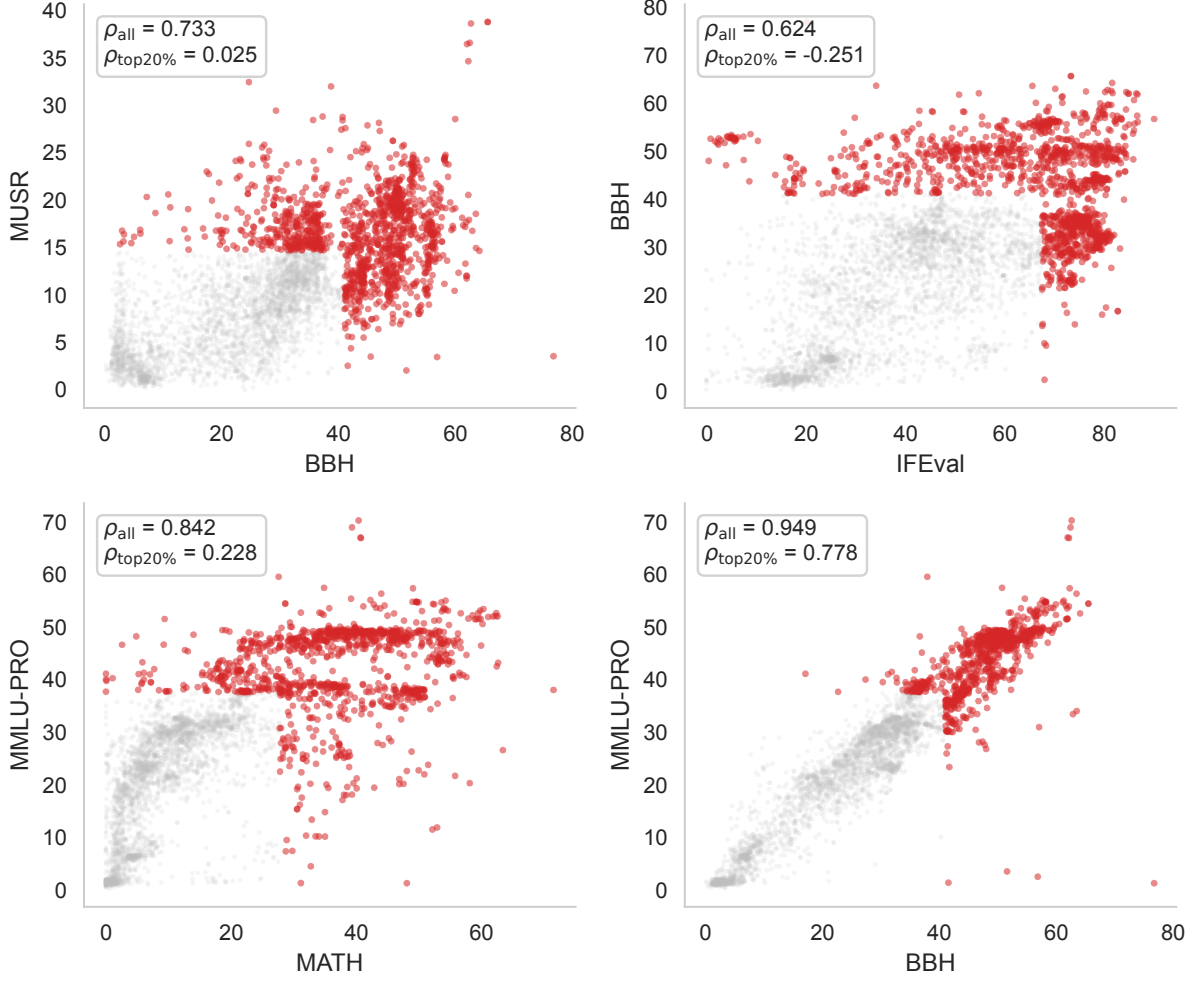


Figure 1: Scatter plots illustrating the conditional Spearman collapse. Gray: all 4,497 models. Red/orange: models in the top 20% on at least one of the two benchmarks (the conditional Spearman ρ_{up} is computed on this same subset—models above the 80th percentile on *either* benchmark). Three deceptive pairs (top row and bottom-left) show high bulk correlation but scatter widely in the tail; the non-deceptive pair (bottom-right) maintains structure.

Whitney $p = 0.95$), confirming no scalar measure detects the tail decomposition.

4.5 Multi-Dataset Validation

Having established the correlation mirage on OLLM v2’s six benchmarks, we test whether it reflects a broader pattern by applying the identical pipeline to two independent datasets.

kensho/WILD (65 models $\times$ 27 benchmarks). Applying the identical pipeline to 351 pairs from an independent evaluation framework ([Kensho Technologies, 2024](#)), Frank copulas again dominate (133/351, 38%), replicating the OLLM v2 pattern. The conditional Spearman collapse repli-

cates decisively (mean drop 58%), with sign reversals such as DROP $\times$ GSM8K ($\rho_S = 0.858 \rightarrow \rho_{\text{up}} = -0.412$). At $n = 65$, individual family assignments carry wider CIs, so WILD provides *directional replication*; pair-level results in Appendix K.

BenchPress (188 models $\times$ 316 benchmarks). On the BenchPress matrix, we replicate the core finding with a sensitivity analysis by sample size. At the recommended minimum threshold of $n \geq 35$ (188 pairs), 55.9% of pairs show $\Delta\text{BIC} > 2$, exceeding the simulation FPR by +10.6 percentage points—demonstrating genuine non-Gaussian dependence (Table 3):

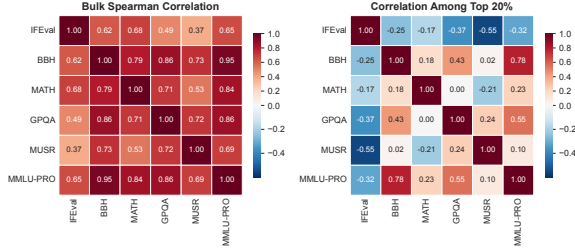


Figure 2: Bulk Spearman (left) vs. conditional Spearman in the upper 20% (right). The dramatic lightening of the right heatmap reveals the collapse.

Min n	Pairs	% $\Delta\text{BIC} > 2$	Excess over FPR
25	398	49.0%	+0.5pp
35	188	55.9%	+10.6pp
50	100	57.0%	+16.2pp
75	31	61.3%	+28.0pp

Table 3: Sensitivity analysis on BenchPress. At $n \geq 25$, the non-Gaussian rate barely exceeds the simulation FPR (+0.5pp), indicating insufficient power. **We recommend $n \geq 35$ as the minimum defensible threshold** (bold row): excess over FPR jumps to +10.6pp, and the signal strengthens monotonically through $n \geq 75$ (+28pp; Figure 5 in Appendix), consistent with copula estimation requiring moderate sample sizes.

The finding that “deceptive redundancy” is real—not a small-sample artifact—is further confirmed by the OLLM v2 results at $n = 4,497$, where the evidence is orders of magnitude stronger.

4.6 Vine Copula: BBH as Tail Information Hub

An R-vine copula fitted to the 6-benchmark OLLM v2 matrix (BIC $-29,198$) identifies **BBH** as the hub node (degree 4 in Tree 1): two of four BBH edges are Frank ($\lambda_U = 0$), while two retain tail structure (BBH–MATH: BB1; BBH–MMLU-Pro: Student-t). PCA selects MMLU-Pro as most central by linear variance; the vine selects BBH by tail structure—a PCA-based pruning strategy would discard the tail-information linchpin (bootstrap stability $\geq 94.9\%$; Figure 3; Appendix B).

5 Related Work

Benchmark Redundancy and Evaluation Methodology. Sha and Zhao (2026) find that the Open LLM Leaderboard v2’s six benchmarks collapse to effective dimensionality ≈ 1.7 ; Subramani et al. (2025) and Zhang et al. (2025) propose correlation-based redundancy criteria;

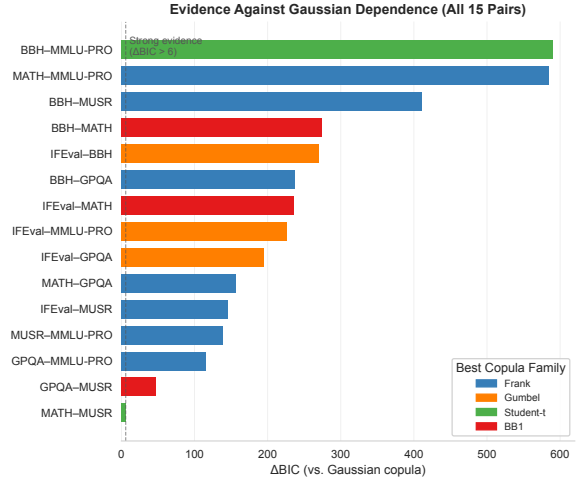


Figure 3: ΔBIC evidence against Gaussian dependence for all 15 OLLM v2 pairs, colored by best-fitting copula family. Fourteen of 15 pairs exceed the strong-evidence threshold ($\Delta\text{BIC} > 6$, dashed line). Frank pairs (blue) dominate, including all three headline deceptive pairs with $\Delta\text{BIC} > 150$. The R-vine structure (BBH as tail information hub, degree 4; Appendix B) is learned from these same non-Gaussian fits.

Kipnis et al. (2025) show via factor analysis that a single factor explains $\sim 79\%$ of variance—complementary to our finding that tail structure diverges from this bulk picture. Polo et al. (2024), Dehghani et al. (2021), Kiela et al. (2021), Liang et al. (2023), and Chiang et al. (2024) address related evaluation methodology concerns. All use linear or monotonic association; none consider tail dependence.

Dependence Beyond Correlation. Correlation breakdown at the extremes is well-established in finance (Longin and Solnik, 2001; Ang and Chen, 2002); Embrechts et al. (2002) formalized this via copulas, with comprehensive treatments by Nelsen (2006); Joe (2014); Czado (2019) and conditional extensions by Patton (2006). Aich et al. (2026) apply copulas to medical feature selection; Safaai (2026) develop amortized vine copulas. To our knowledge, no prior work applies copulas to NLP benchmark evaluation. IRT-based approaches (Lalor et al., 2019; Rodriguez et al., 2021) model item-level informativeness at different ability levels—a suggestive single-benchmark analogue; our framework addresses the complementary *pairwise* question. Ethayarajh and Jurafsky (2020) raise related evaluation concerns.

6 Why Does Correlation Collapse?

Three alternative explanations could produce the conditional Spearman collapse; we address and rule out each.

Not a ceiling effect or measurement noise. Among the top-20% BBH models ($n = 900$), GPQA scores span the *entire* range (spread ratio = 1.00); MUSR retains 95% of its spread. The benchmarks are not compressed at the top; they *diverge*. A measurement-noise explanation (binomial ceiling noise attenuating all correlations) is inconsistent with the *differential* pattern: BBH×MMLU-Pro maintains $\rho_{\text{up}} = 0.778$ ($\hat{\lambda}_{5\%} = 0.61$) while BBH×MUSR collapses to $\rho_{\text{up}} = 0.025$ ($\hat{\lambda}_{5\%} = 0.16$). Ceiling noise would attenuate all pairs similarly; the dichotomy demonstrates genuine capability trade-offs.

Not Berkson’s paradox or range restriction. Calibrating against four null models (Gaussian, Student-t $\text{df} = 5, 10, 30$, permutation) at matched (n, ρ) , all headline pairs exceed every null at the 95th percentile (BBH×GPQA: excess = -0.55 , $p < 0.001$; BBH×MUSR: -0.51 ; MATH×MMLU-Pro: -0.59). Critically, excesses against the Student-t null exceed those against Gaussian, because the Student-t predicts more tail correlation—making the observed attenuation even more anomalous (Appendix G).

Consistent with capability specialization. The collapse is *consistent with* mid-range models sharing architectural constraints (bulk correlation) while frontier models diverge through specialization; the mechanism remains open.

Robustness. All findings survive deduplication to 779 base-model families ($r = 0.96$; cluster bootstrap stability 98%/96% for BBH×MUSR/MATH×MMLU-Pro; Appendix I), expanded copula candidate set (8→30+ families; Frank or tail-equivalent; Appendix D), and threshold sensitivity (11/15 pairs identical at 70th–95th percentiles; Appendix H).

7 Practical Recommendations

We propose a concrete **three-step protocol** for benchmark suite maintainers that requires minimal statistical expertise:

Step 1: Screen. Compute pairwise Spearman ρ_S as usual. Identify candidate “redundant” pairs

with $\rho_S > 0.7$.

Step 2: Check the tail. For each candidate pair, compute conditional Spearman restricted to models in the upper performance region. A simple pairwise implementation for benchmarks A and B:

```
mask = (df[['A', 'B']].rank(pct=True) >
        0.8).any(axis=1)
df.loc[mask,
        ['A', 'B']].corr(method='spearman')
```

This selects models above the 80th percentile on *either* benchmark (union conditioning, matching the paper’s ρ_{up} definition). If the resulting correlation $< 0.5 \cdot \rho_S$, flag the pair as potentially deceptive.

Step 3: Decide. For flagged pairs, retain both benchmarks or run the copula analysis; conditional Spearman at the 80th percentile captures the essential phenomenon.

Step 3b: Diagnose reliability vs. capability specialization. If noise is suspected, check whether the benchmark’s *other* pairings also collapse: uniform collapse across all pairings suggests noise; selective collapse—e.g., BBH×MMLU-Pro retains $\rho_{\text{up}} = 0.778$ while BBH×MUSR collapses to 0.025 under identical conditioning—points to genuine capability specialization.

Key rule: Never remove a benchmark solely because $\rho_S > 0.8$; run the three-step protocol above and report conditional Spearman heatmaps or λ_U alongside standard correlation matrices. Full pipeline released as supplementary code.

8 Conclusion

Seven of 15 OLLM v2 pairs exhibit Frank copulas with zero asymptotic tail dependence despite ρ_S up to 0.86—pairs that appear redundant by every standard criterion yet provide non-redundant ranking information precisely where practitioners make their most consequential comparisons: at the frontier. The conditional Spearman collapse (50–97%, with sign reversals at the top 5%) survives deduplication to 779 base-model families ($\geq 96\%$ bootstrap stability) and is independently replicated on BenchPress and kensho/WILD (38–47% Frank prevalence, 58% mean collapse); dropping a benchmark because $\rho_S > 0.8$ discards the signal that separates elite models. Our three-step protocol and released toolkit make the corrective practical; the framework anticipates similar divergence in multimodal evaluation suites.

Limitations

(1) OLLM v2 has only 6 benchmarks; the BenchPress analysis (316 benchmarks) provides breadth but at smaller n . (2) The $\lambda_U = 0$ claim is parametric (a property of the Frank copula family); our nonparametric $\hat{\lambda}_{5\%}$ confirms the trend but cannot distinguish “exactly zero” from “small positive” tail dependence at finite n . (3) The OLLM v2 benchmarks are all reasoning/knowledge-focused; multimodal or generation benchmarks may behave differently. (4) OLLM v2’s many fine-tuned variants overstate effective degrees of freedom and raise a relevance concern for practitioners focused solely on base-model selection. We address both: statistically, deduplication to 779 base-model families preserves headline findings (cluster bootstrap stability 96%–98% for BBH×MUSR and MATH×MMLU-Pro); substantively, BenchPress (316 diverse benchmarks) and kensho/WILD (independent organizations, 27 benchmarks) independently replicate the Frank copula prevalence (38–47%), demonstrating the finding is not an artifact of OLLM v2’s composition. Fine-tune and merge selection is itself a real practitioner decision (e.g., quantized-variant selection for production), and the tail-aware protocol applies there directly. (5) The exact mechanism (capability specialization vs. training data composition) remains an open question. (6) An alternative explanation invokes classical test theory: near-ceiling performance reduces discriminating items, and binomial sampling noise mechanically attenuates correlations. Two observations argue against this as a *sole* cause: (a) the collapse is *differential* (BBH×MMLU-Pro retains $\rho_{\text{up}} = 0.78$ while BBH×MUSR collapses to 0.025 under identical conditioning)—noise on the shared BBH marginal would attenuate both pairs; (b) headline pairs exceed Student-t null models that already account for noise-driven attenuation ($p < 0.001$; Appendix G). Fully disentangling capability specialization from measurement noise remains an open direction (Kiel et al., 2021).

Ethical Considerations

This work involves purely statistical analysis of publicly available benchmark scores from the Open LLM Leaderboard v2 and BenchPress. No human subjects, private data, or model inference were involved. All datasets used are publicly accessible and have been previously released under

open licenses. Our findings aim to improve evaluation transparency by revealing that correlation-based redundancy metrics can obscure meaningful distinctions among top-performing models. We note that tail-aware benchmark analysis could, in principle, be used to identify which benchmarks are most informative at the frontier and thus concentrate evaluation effort; we view this as a positive application that reduces wasteful evaluation without sacrificing discriminative power. We are not aware of negative societal impacts specific to this work beyond those inherent to LLM evaluation research generally.

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A Complete Bivariate Copula Fit Results

The complete set of 15 bivariate copula fits for the OLLM v2 dataset ($n = 4,497$) and the 398 pairs for the BenchPress dataset are provided in the supplementary `bivariate_fits_pairwise.json` file. For each benchmark pair, we report: best-fitting copula family (by BIC), estimated parameters, Spearman ρ , Kendall τ , λ_U , λ_L , and tail asymmetry $|\lambda_U - \lambda_L|$.

B Vine Copula Details

The R-vine copula was fitted using `pyvinecopulib` v0.7.6 with automatic structure selection (log-likelihood 14,683; Akaike Information Criterion (AIC) $-29,326$; BIC $-29,198$). The vine matrix, pair-copula families at each tree level, and estimated parameters are provided in `vine_structure.json` and `ollm_vine_structure.json`.

Tree 1 structure. BBH serves as the hub (degree 4) with edges: BBH-GPQA (Frank, $\lambda_U = 0$), BBH-MUSR (Frank, $\lambda_U = 0$), BBH-MATH (BB1, $\lambda_U = 0.65$), BBH-MMLU-Pro (Student-t, $\lambda_U = 0.66$). The remaining edge connects IFEval to MMLU-Pro via Frank ($\lambda_U = 0$). Three of five Tree 1 edges have zero asymptotic tail dependence; the two with positive λ_U (BBH-MATH and BBH-MMLU-Pro) correspond to the non-Frank pairs in Table 1.

The BBH-MMLU-Pro assignment (Student-t, $\lambda_U = 0.665$, 99.6% bootstrap stability) is consistent with Table 1.

Bootstrap stability of BBH edges. Bivariate bootstrap (1,000 resamples) for the four BBH edges yields high family stability: BBH-GPQA 100% (Frank), BBH-MUSR 100% (Frank), BBH-MATH 94.9% (BB1), BBH-MMLU-Pro 99.6% (Student-t).

Comparison with PCA centrality. PCA-based centrality (sum of absolute loadings on PC1-PC3): MMLU-Pro ranks highest, followed by BBH. Vine copula centrality (Tree 1 degree): BBH ranks highest (degree 4), MMLU-Pro is peripheral (degree 2). The disagreement reflects the distinction between linear variance structure (PCA) and tail dependence structure (vine).

Pair	Original	Expanded	ΔBIC
IFEval $\times$ BBH	Gumbel	Tawn (rot=180)	+294
IFEval $\times$ MATH	BB1	Tawn (rot=180)	+222
IFEval $\times$ MMLU-Pro	Gumbel	BB8 (rot=180)	+390
BBH $\times$ GPQA	Frank	BB8	+97
BBH $\times$ MUSR	Frank	Frank	0
BBH $\times$ MMLU-Pro	Student-t	Student-t	0
MATH $\times$ GPQA	Frank	Frank	0
MATH $\times$ MMLU-Pro	Frank	Frank	0
MUSR $\times$ MMLU-Pro	Frank	Frank	0

Table 4: Family assignment: original (8 families) vs. expanded (11 + rotations). Pairs not shown: IFEval $\times$ GPQA (Gumbel $\rightarrow$ BB8 rot=180, $\Delta\text{BIC}=+107$), IFEval $\times$ MUSR (Frank $\rightarrow$ BB8 rot=180, +57), BBH $\times$ MATH (BB1 $\rightarrow$ Tawn rot=180, +99), MATH $\times$ MUSR (Student-t $\rightarrow$ BB8, +167), GPQA $\times$ MUSR (BB1 $\rightarrow$ BB8, +75), GPQA $\times$ MMLU-Pro (Frank $\rightarrow$ BB8, +98). The four headline Frank pairs (BBH $\times$ MUSR, MATH $\times$ GPQA, MATH $\times$ MMLU-Pro, MUSR $\times$ MMLU-Pro) remain Frank.

C Marginal Distributions

Raw score histograms and PIT-transformed densities for all 6 benchmarks in the OLLM v2 dataset are provided in the supplementary material (`marginals_and_pit.pdf`). All benchmarks pass the Kolmogorov-Smirnov uniformity test after rank-based probability integral transform (PIT) ($p > 0.94$).

D Expanded Copula Candidate Set

To verify robustness to the choice of copula candidate set, we expand from 8 to 11 families (adding BB6, BB8, Tawn) and enable rotations (90°, 180°, 270°) of all asymmetric families, yielding over 30 candidate copulas per pair.

Tail independence is invariant to family shift. The shift from Frank to BB8 for BBH $\times$ GPQA does not represent a failure of stability; it represents a closer parametric fit to the bulk interior. The BB8 (Joe-Frank) copula is a two-parameter family whose upper tail dependence coefficient is:

$$\lambda_U = \begin{cases} 2 - 2^{1/\theta} & \text{if } \delta = 1 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Because the estimated parameters for BBH $\times$ GPQA yield $\delta < 1$, the BB8 copula retains $\lambda_U = 0$ exactly. The physical property of interest—asymptotic tail independence—is thus fully preserved under candidate set expansion: the family label changes but the tail classification does not. Similarly, for GPQA $\times$ MMLU-Pro

(Frank→BB8, +98), the estimated BB8 parameters also yield $\lambda_U = 0$. This confirms that the “100% bootstrap stability” of Frank under the 8-family set reflects a genuine structural property (zero tail dependence) rather than an artifact of a restricted candidate library.

E Student-t Degrees of Freedom

When fitting Student-t copulas to all 15 pairs, the estimated degrees of freedom range from 5.0 (BBH×GPQA) to 36.1 (MUSR×MMLU-Pro), with a median of 7.7. Ten of 15 pairs have $df < 10$, indicating heavy-tailed dependence structure. This motivates the Student-t null in the Berkson calibration (§6).

F Power Analysis

At $n = 779$ (deduplicated sample), BIC-based copula selection achieves >99% power to detect $\lambda_U \geq 0.05$ (Gumbel copula with matching θ), across 500 simulations per condition. This means the Frank copula assignment ($\lambda_U = 0$) for our headline pairs is not driven by insufficient power—the procedure can reliably distinguish even small positive tail dependence from zero.

G Broadened Berkson Calibration

We calibrate the conditional Spearman collapse against four null models: Gaussian copula, Student-t copula ($df = 5, 10, 30$), and a permutation null. The excess beyond each null is computed at the 80th, 90th, and 95th percentiles. Complete results for all 15 pairs are provided in `berkson_calibration_broadened.json`. The headline finding: all three core deceptive pairs (BBH×GPQA, BBH×MUSR, MATH×MMLU-Pro) show statistically significant excess ($p < 0.001$) beyond the most conservative Student-t($df = 5$) null at the 90th and 95th percentiles.

H Threshold Sensitivity Ablation

Conditional Spearman is computed at the 70th, 75th, 80th, 85th, 90th, and 95th percentiles for all 15 pairs. Of 15 pairs, 11 receive identical deceptive/non-deceptive classification at all six thresholds. Zero classification changes occur between the 75th and 80th percentile. Effective sample sizes at the 80th percentile range

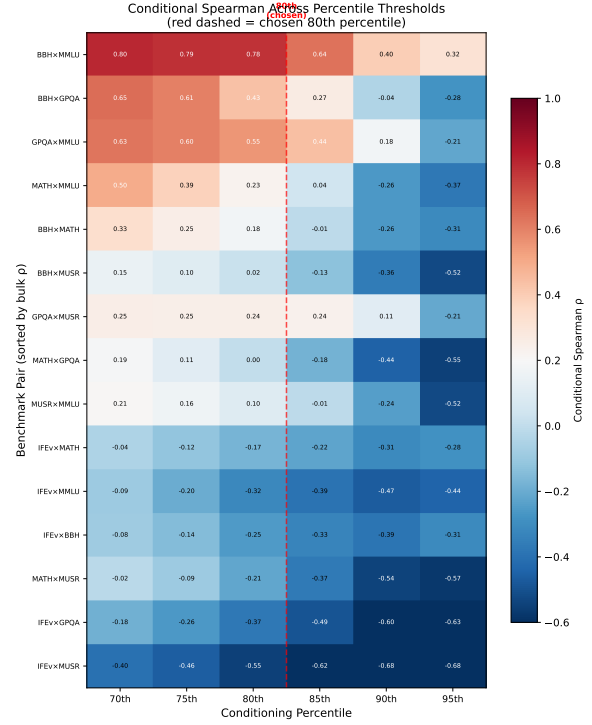


Figure 4: Threshold sensitivity ablation. Conditional Spearman for all 15 pairs across the 70th–95th percentiles (pairs sorted by bulk ρ_S). Of 15 pairs, 11 receive identical deceptive/non-deceptive classifications at all thresholds. The red dashed line marks the chosen 80th percentile, which lies on a stability plateau (zero classification changes from 75th to 80th). Dark red cells indicate sign reversal.

from 1,026 to 1,539 across pairs. Full results: `threshold_sensitivity_ablation.json`.

I Cluster Bootstrap at $n = 779$

To account for non-independence among fine-tuned model variants, we perform cluster bootstrap (1,000 resamples) at the base-model family level (714 unique clusters) on the deduplicated $n = 779$ dataset. Family stability for headline pairs: BBH×MUSR 98% (Frank), MATH×MMLU-Pro 96% (Frank), BBH×GPQA 76% (Frank). Δ BIC confidence intervals: BBH×MUSR [3.7, 71.5], MATH×MMLU-Pro [36.9, 139.8], BBH×GPQA [−10.5, 78.0].

Interpretation. Two of three headline pairs (BBH×MUSR and MATH×MMLU-Pro) show robust non-Gaussian structure after deduplication, with Δ BIC CIs entirely above zero. For BBH×GPQA, the 95% CI spans zero, meaning the Gaussian null cannot be rejected for this spe-

Symbol	Definition
ρ_S	Spearman rank correlation (bulk, all n models)
$\rho_{\text{up}}(q)$	Conditional Spearman: Spearman rank correlation restricted to models above the q th percentile on <i>either</i> benchmark ($U_1 \geq q$ or $U_2 \geq q$); default $q = 0.80$
$\rho_{\text{up}}^{\cap}(q)$	Intersection-conditioned Spearman: restricted to models above threshold on <i>both</i> benchmarks ($U_1 \geq q$ and $U_2 \geq q$); used to verify robustness to union-conditioning collider concerns
λ_U	Upper tail dependence coefficient (asymptotic; from fitted copula family)
λ_L	Lower tail dependence coefficient (asymptotic)
$\hat{\lambda}(k)$	Nonparametric empirical tail concentration: fraction of models jointly in the top- $k\%$ on both benchmarks, divided by k
ΔBIC	$\text{BIC}_{\text{Gaussian}} - \text{BIC}_{\text{best non-Gaussian}}$; positive values favor non-Gaussian dependence
τ	Kendall’s rank correlation coefficient
n_{eff}	Effective sample size after conditioning (union of both marginals: 314–437 at the 5% threshold, 1,026–1,539 at 20%)

Table 5: Notation used throughout the paper. The conditioning scheme for ρ_{up} selects models above the q th percentile on *either* benchmark ($U_1 \geq q$ or $U_2 \geq q$), ensuring adequate sample sizes for reliable Spearman estimation. At the 95th percentile (top 5%), n_{eff} ranges from 314 (high- ρ pairs) to 437 (low- ρ pairs) across the 15 pairs (mean 385); at the 80th percentile, from 1,026 to 1,539.

cific pair at the 95% level after accounting for model-family non-independence. This is consistent with its lower family stability (76%) and indicates that BBH×GPQA’s non-Gaussian structure is statistically weaker than the other two headline pairs. The conditional Spearman collapse (the paper’s primary diagnostic) remains significant for all three pairs regardless. Complete dual- n table: `deduplicated_inference.json`.

J Notation Reference

Table 5 summarizes the key symbols used throughout the paper.

K kensho/WILD Validation Details

The kensho/WILD dataset comprises 65 models evaluated on 27 benchmarks with 100% fill rate, yielding 351 bivariate pairs. We apply the identical 8-family BIC copula selection pipeline used

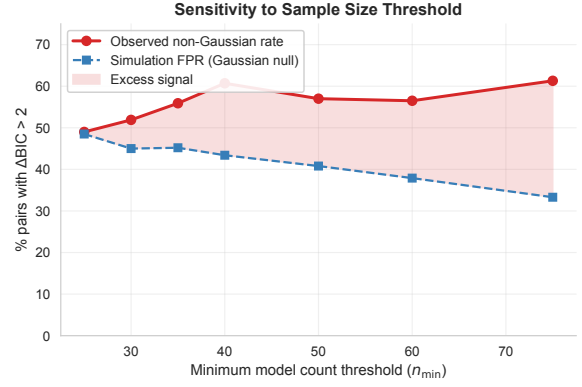


Figure 5: Sensitivity analysis on the BenchPress dataset. The proportion of non-Gaussian pairs ($\Delta\text{BIC} > 2$) grows monotonically with the minimum sample size threshold, confirming the phenomenon scales.

for OLLM v2.

Family distribution. Frank: 133/351 (38%); Gumbel: 104 (30%); Gaussian: 67 (19%); Clayton: 34 (10%); Joe: 9 (3%); Student-t: 4 (1%). The Frank prevalence (38%) is consistent with OLLM v2 (47%), accounting for the lower statistical power at $n = 65$.

Deceptive pairs. Of 351 pairs, 144 (41%) meet the deceptive criteria ($\rho_S > 0.7$ and $\lambda_{\max} < 0.05$). Top examples: ARC-Easy×HellaSwag ($\rho_S = 0.963$, Frank, $\Delta\text{BIC} = 2.7$), GSM8K×MMLU-Pro ($\rho_S = 0.950$, Frank, $\Delta\text{BIC} = 0.8$).

Conditional Spearman collapse. The mean conditional Spearman drop in the upper 20% is 0.58 (median 0.54). The effective sample size per tail conditioning ($n_{\text{eff}} \approx 17$ –21) is small; individual estimates carry standard error $\text{SE}(\rho) \approx 0.22$ –0.25. Despite this, the *aggregate* pattern is unambiguous: of 351 pairs with sufficient tail observations, 349 (99.4%) show positive drops (full ρ exceeds conditional ρ), matching OLLM v2’s universal attenuation pattern.

Complete pair-level results: `wild.bivariate_fits.json`.

L BenchPress Sensitivity Figure

The monotonic increase from 49.0% ($n \geq 25$) to 61.3% ($n \geq 75$) demonstrates that non-Gaussian dependence is a genuine feature of benchmark data rather than a small-sample artifact: larger samples provide more power to detect the departure from Gaussianity.

M Reproducibility

All code and data processing scripts are provided in the supplementary material. The analysis requires only publicly available data from the Open LLM Leaderboard v2, the BenchPress score matrix ([Zeng, 2026](#)) (dataset URL provided in supplementary README), and standard Python libraries (`pyvinecopulib`, `scipy`, `pandas`, `scikit-learn`).

No model inference was performed. Total computation time on a standard laptop: under 5 minutes (including all robustness analyses).